

A c_0 counterexample for sequential weak lower semicontinuity of Pasch–Hausdorff envelopes

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Status: claimed full counterexample, likely valid, pending human review. The counterexample works for every envelope parameter $k > 0$.

1 Source question

Tang, Zhang, and Guo [1] ask whether Pasch–Hausdorff infimal convolution preserves sequential weak lower semicontinuity on an arbitrary Banach space. Their precise question appears in Section 2 of arXiv:2110.05257v1, printed/PDF page 10. They prove an affirmative result for reflexive spaces (Theorem 2.8), observe that it also holds in Schur spaces (Remark 2.9), and predict that the general answer is negative.

the lower semicontinuity of f if f is l.s.c and bounded below. Since the notion of a sequentially weakly lower semicontinuity is stronger than that of a lower semicontinuity, a natural problem is whether f_k can also retain the sequentially weakly lower semicontinuity of f if f is s.w.l.s.c and bounded below. Currently, we only have a partial answer to the problem although we think that the final answer is “No”. Theorem 2.8 below shows that the answer is Yes when the underlying space is reflexive, whereas the following Remark 2.9 shows that the reflexivity is not necessary.

Figure 1: The open question in arXiv:2110.05257v1, page 10.

For a proper function $f : E \rightarrow (-\infty, +\infty]$ bounded below and $k > 0$, the Pasch–Hausdorff envelope is

$$f_k(x) = \inf_{y \in E} (f(y) + k\|y - x\|).$$

The question asks whether f_k must be sequentially weakly lower semicontinuous whenever f is.

2 Idea of the counterexample

For an indicator function $f = I_A$, the envelope is simply the distance function $f_k = k d(\cdot, A)$. Thus one needs a sequentially weakly closed set A , a weakly null sequence (x_n) , and a strict gap $d(x_n, A) < d(0, A)$. In c_0 , the vectors

$$y_n = e_n + c(e_1 + \cdots + e_{n-1})$$

have norm one but follow, coordinate by coordinate, the constant vector $(c, c, \dots)$, which lies outside c_0 . Consequently their set has no nontrivial weakly convergent sequence and is sequentially weakly closed. At the same time, y_n lies within distance c of the weakly null vector e_n .

3 Counterexample and proof

Theorem 1. *Let $E = c_0$ over the real scalars and fix any $c \in (0, 1)$. Define*

$$y_n = e_n + c \sum_{j=1}^{n-1} e_j, \quad A = \{y_n : n \in \mathbb{N}\},$$

and let $f = I_A : E \rightarrow [0, +\infty]$. Then f is proper, bounded below, and sequentially weakly lower semicontinuous. Nevertheless, for every $k > 0$, f_k is not sequentially weakly lower semicontinuous at the origin.

Proof. We first prove that A is sequentially weakly closed. Suppose a sequence (y_{n_m}) in A converges weakly to some $y \in c_0$. If (n_m) is unbounded, pass to a strictly increasing subsequence. For every fixed coordinate j , one then has

$$y_{n_m}(j) = c$$

for all sufficiently large m . Since coordinate evaluations belong to $c_0^* = \ell_1$, weak convergence would force $y(j) = c$ for every j . This is impossible because the constant sequence $(c, c, \dots)$ does not belong to c_0 . Thus (n_m) is bounded. A convergent sequence taking values in a finite Hausdorff set is eventually constant, so its limit belongs to A .

The indicator of a sequentially weakly closed set is sequentially weakly lower semicontinuous. Hence $f = I_A$ has all the properties asserted in the first sentence. Directly from the definition,

$$f_k(x) = \inf_{y \in A} k \|y - x\| = k d(x, A). \quad (1)$$

For every n , $\|y_n\|_\infty = 1$, and therefore

$$d(0, A) = 1, \quad f_k(0) = k.$$

On the other hand,

$$\|e_n - y_n\|_\infty = \left\| c \sum_{j=1}^{n-1} e_j \right\|_\infty \leq c,$$

so $f_k(e_n) \leq kc$. The canonical basis (e_n) is weakly null in c_0 : for $a = (a_j) \in \ell_1 = c_0^*$, one has $\langle a, e_n \rangle = a_n \rightarrow 0$. Consequently,

$$\liminf_{n \rightarrow \infty} f_k(e_n) \leq kc < k = f_k(0),$$

which violates sequential weak lower semicontinuity at zero. □

4 Verification, scope, and novelty

Every hypothesis is used literally: f is extended-real-valued, proper, bounded below by zero, and nonconvex; the source question explicitly permits this class. The same construction works over the complex scalars. It does not contradict the source's positive results for reflexive or Schur spaces, since c_0 has neither property.

The novelty check was bounded as follows. The run registry, solution, attempt, and proof-gap indexes were searched for arXiv:2110.05257, Pasch–Hausdorff envelopes, infimal convolution, and sequential weak lower semicontinuity. The full local source corpus was searched for the exact question

phrase. Exact-phrase and arXiv-focused web searches combined these terms with “counterexample” and “ c_0 ”. They found the source and general envelope literature but no later answer or matching construction. Novelty confidence is therefore moderate rather than certified.

The recommended human check is the sequential weak closedness of A ; after that, the identity (1) and the strict distance gap are immediate.

References

- [1] Y. Tang, S. Zhang, and T. Guo, *Some nonlinear characterizations of reflexive Banach spaces*, arXiv:2110.05257v1 (2021); J. Math. Anal. Appl. 519 (2023), 126736.